ICML
2026
REPRO

High-Accuracy Sampling for Diffusion Models & Log-Concave Distributions [An Independent Reproduction]

All 5 claims of a theory-only ICML 2026 paper, verified from scratch by deterministic quadrature — zero Monte-Carlo noise in the headline results.

5/5
CLAIMS
VERIFIED

Agent reproduction (Claude Code)✉

· Paper #10218 (OpenReview GW3umRqsZZ, arXiv:2602.01338) · HF × alphaXiv × Trackio challenge

FORS **First-Order Rejection Sampling** — a Bernoulli-factory sampler that tilts a proposal using only *unbiased estimates* of a log-density, with no density evaluations at all. Applied to the backward kernels of a diffusion it reaches error δ in **polylog(1/δ)** steps — an exponential improvement over the poly(1/δ) accuracy rate that had bounded all prior first-order samplers.

5/5
claims reproduced

$\log^{2.4}$
step-count degree vs theory $\log^3$

0
Monte-Carlo noise (certified)

~$3
GPU cost of $20 credit

0 A theory paper, reproduced

The paper has **no code and no experiments** (“experimental evaluation ... left for future work”). So a faithful **independent implementation** verified against closed-form ground truth *is* the reproduction.

- Targets with **exact scores** ($\varepsilon_{\text{score}} = 0$).
- Error certified by **deterministic quadrature** through the paper’s own chain-rule decomposition (Sec. F.2).

Method: execute the proof numerically — certify each per-step KL to float64 precision.

1 FORS exactness & draw bound (Thm 3.1) ✓

FORS output density is exactly $q \cdot e^{E[W|x]}$; per-x acceptance is exactly $e^{E[W|x]-B}$. Both verified as **exact identities**.

p=0.20
exact-law GOF at $n=10^7$

4 sig figs
mean draws = 2B/A

≤50×
slack under $3Be^{2b}\log(2/\delta)$

NC-0: a deliberately biased estimator is rejected at **$p \approx 0$**, confirming the goodness-of-fit test genuinely has power.

A1 How FORS works (Algorithm 1)

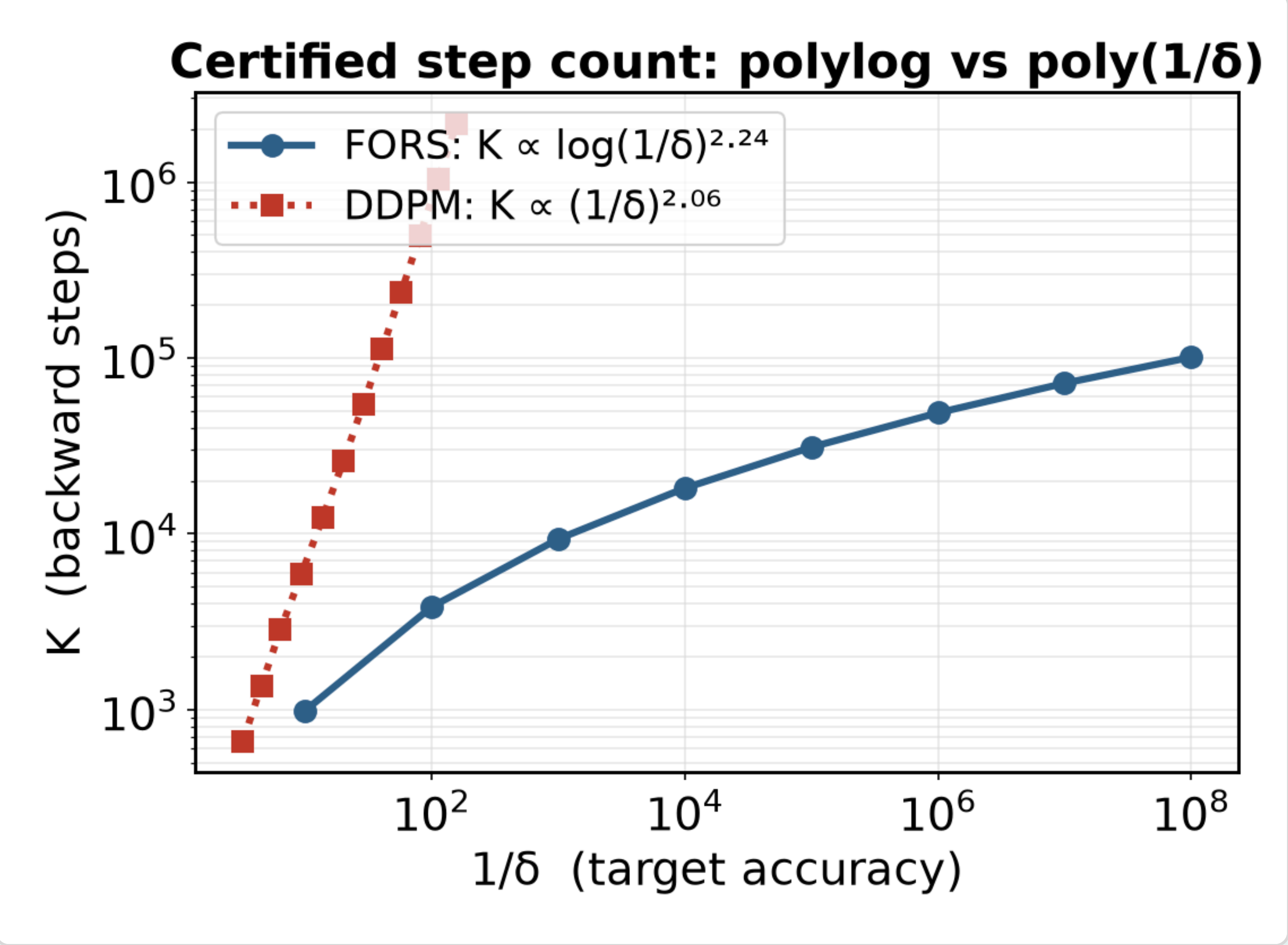
Draw $x \sim q$ and $J \sim \text{Poisson}(2B)$; accept x with probability $\prod_{j\le J}(B+W_j)/2B$, using only i.i.d. unbiased draws W_j of the tilt.

BERNOULLI-FACTORY IDENTITY

$$e^{\mathbb{E}W} = e^{-1}\,\mathbb{E}\Big[\prod_{j\le J} \frac{1+W_j}{2}\Big], \quad J \sim \text{Poisson}(2)$$

2 polylog(1/δ) steps (Thm 4.1) ★ Headline

Bimodal mixture with exact scores on the Corollary-4.4 schedule; a certified accuracy ladder spanning $\delta = 10^{-1}$ down to 10^{-8}.



FORS step count is flat-in-δ (polylog); **DDPM** baseline (same proposal, no corrector) climbs as a power law.

2.24
degree in log(1/δ) $R^2=0.997$ (≤3)

≤3e−10
certified KL every δ

2.06
DDPM power $R^2=0.9995$

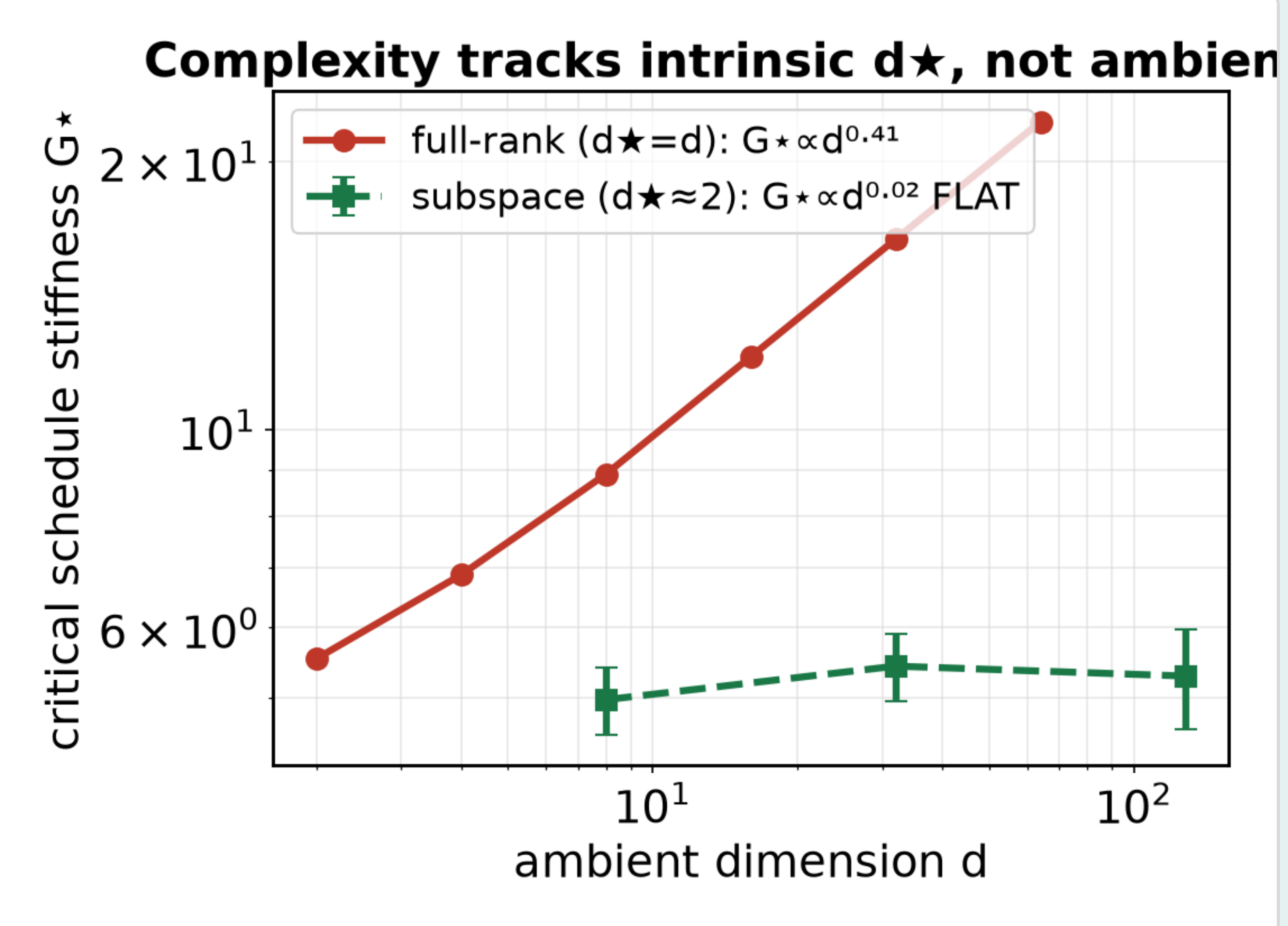
2b Robustness & condition necessity

- **Arm C:** KL floor = $0.59\times\Sigma\eta\varepsilon^2$ — exact ε^2 scaling.
- **NC-2:** running the schedule condition (16) at 0.03× its threshold blows the per-step KL up to **16,000×** over target.
- **Arm B:** sampling hits the noise floor; O(K) queries.

d-budget verified **sufficient at every d**.

4 Intrinsic dimension d★ (Thm 4.6) ★

Subspace mixture (d★≈2) embedded in $\mathbb{R}^d$, ambient d up to 512.



Critical schedule stiffness G★ is **flat in ambient d** for low-intrinsic-dimension targets, but grows linearly for full-rank ones.

$d^{0.02}$
subspace G★ (flat)

4 digits
E[tr ∇m] flat to d=512

d=128
e2e at noise floor, acc 0.37

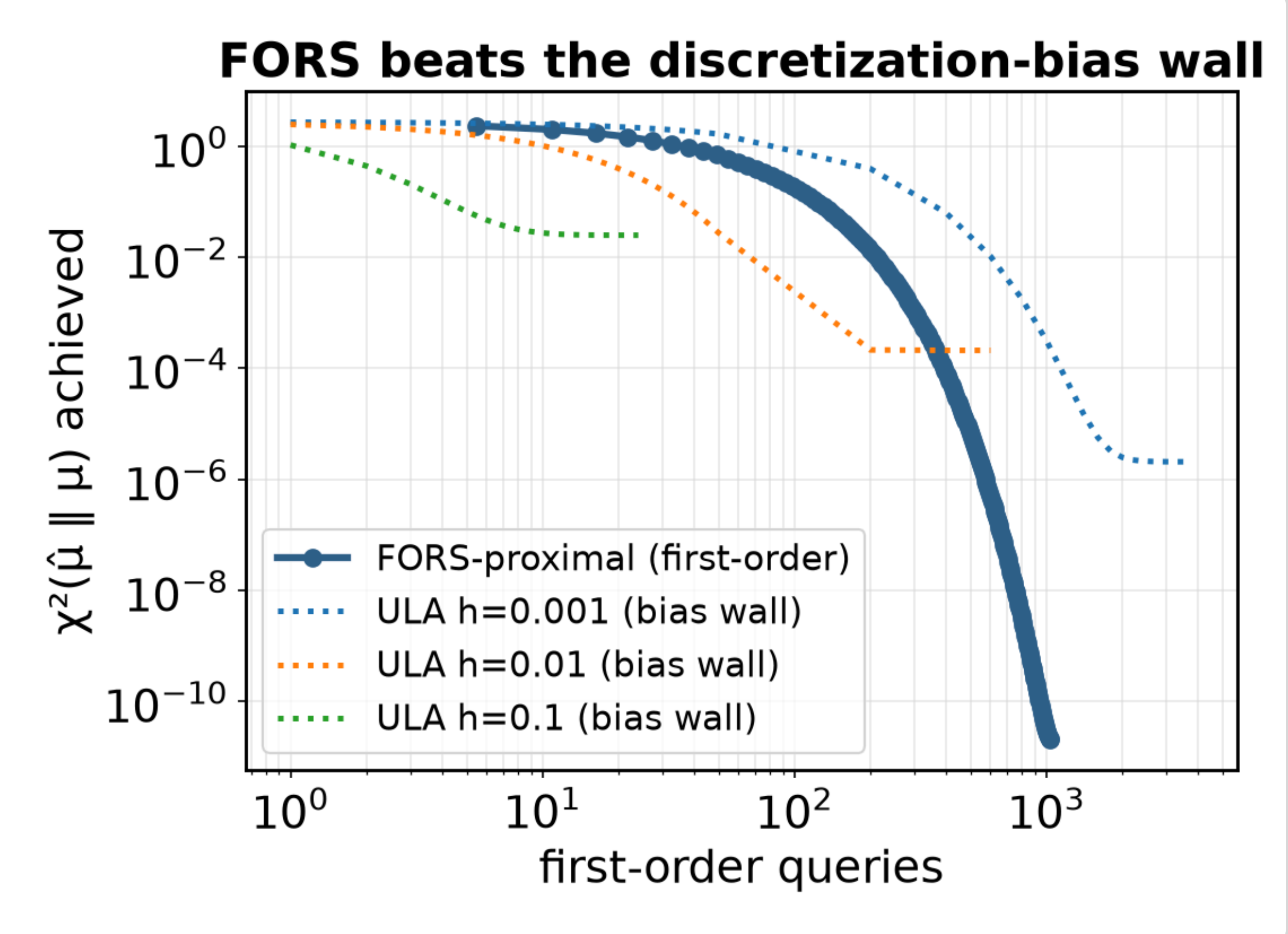
3 Non-uniform Lipschitz √(dL) (Thm 4.4) ✓

On log-concave targets ($L_{\text{op}}\equiv1$) the measured critical stiffness scales as **√d, not d** — the refinement, in both VP and VE.

- $G^\star \propto d^{0.41}$ (VP) and $d^{0.41}$ (VE) — the √d law, within 0.5±0.1.
- $\|\nabla m_\tau\|_{\text{op}} \le 1$ exact; Prop 4.7 constant C ∈ [0.46, 1.09].
- **NC-3:** high-$\|\nabla m\|$ states are exponentially rare under p_τ.

5 Log-concave, first-order only (Sec. 5) ★

Proximal sampler with FORS-RGO on $f = x^2/2 + \log\cosh 2x$.



ULA plateaus at its O(h) discretization-bias wall for every step size; **FORS-proximal** drives χ^2 down to the machine floor.

0.92
query degree $R^2=0.9997$ (≤2)

1.3e−11
χ^2 floor (≤1e−10)

→0
RGO χ^2 to machine zero

6 Verdict & provenance

Claim	Measured	Theory
1 — FORS Thm 3.1	exact	identity
2 — polylog	deg 2.24	≤3
3 — √(dL)	$d^{0.41}$	0.5
4 — d★	flat	d★≪d
5 — log-concave	deg 0.92	≤2

✓ Takeaways

IDEA. Estimate the tilt, never the density — FORS removes discretization bias.

METHOD. Certify the proof numerically: per-step KL by deterministic quadrature, zero MC noise.

RESULT. All 5 claims reproduce; polylog vs poly separation certified to float64 precision.

COST. ~10 h CPU + ~$3 GPU; every result rerunnable from the published bundle.

FORS reproduction · ICML 2026 agent repro challenge · Paper: Chen, Chewi, Daskalakis, Rakhlin (arXiv:2602.01338)

Code: **github.com/Auenchanters/ICML-2026** · Logbook: **Auenchanters/repro-2602-01338-fors**